



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**FINAL EXAMINATION
SEMESTER III, SESSION 2020/2021
PART II**

COURSE CODE	:	FSPC 0034
COURSE NAME	:	CHEMISTRY II
PROGRAMME	:	ASASI UTMSPACE
DURATION	:	1 HOUR 30 MINUTES
DATE	:	15 APRIL 2021

INSTRUCTION TO CANDIDATES:

1. DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DOSO.
2. THERE ARE **FOUR (4)** QUESTIONS. ANSWER ALL QUESTIONS.

WARNING!

Students caught copying/cheating during the examination will be liable for disciplinary actions and UTMSPACE may recommend the student to be expelled from the study.

This booklet consists of (8) printed pages including this page.

QUESTION 5 (10 MARKS)

a) State the oxidation number of the following elements:

i. sulfur in H_2SO_4

$$2(+1) + 1(x) + 4(-2) = 0, x = +6$$

ii. zinc in the compound $\text{Zn}(\text{OH})_2$

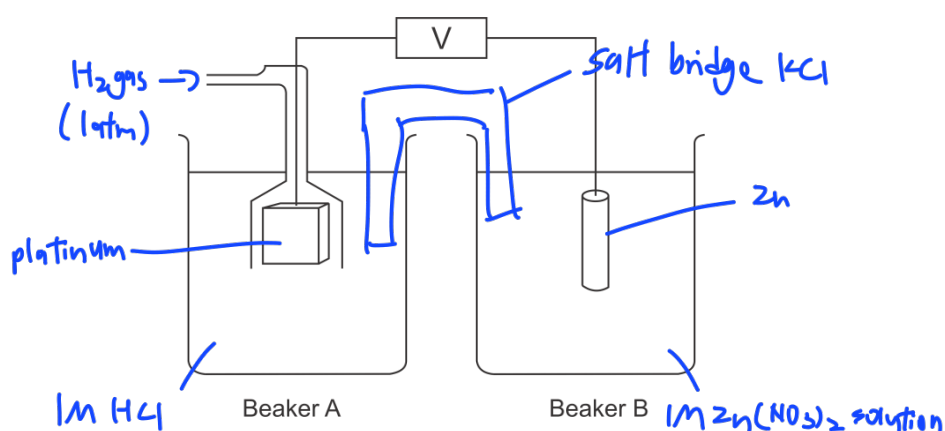
$$x + 4(-2) + 4(+1) = -2$$

$$x - 8 + 4 = -2$$

$$x = +2$$

(1 + 1 marks)

b) The figure below shows a set of apparatus used to determine the standard reduction potential of zinc. The experiment was carried out at 25°C .



i. Suggest the solution to be used in each half-cell. State the concentration of each solution. $\text{Zn}(\text{NO}_3)_2$, 1.0M

ii. The diagram shown is incomplete. Draw and label the apparatus involved.

iii. Write the chemical equation for the overall cell reaction.

(2 + 3 + 3 marks)



QUESTION 6 (10 MARKS)

- a) The car spare parts industry involved chromium plating. This is an example of electrolysis, in which car bumpers are suspended in a dichromate solution in acidic condition.



- i. Write a balanced half reaction at the cathode for the above electrolysis.

- ii. ^{lead anode} State the common anode that is used in this process.

- iii. How long would it take to apply chromium plating 1.0×10^{-2} mm thick to the object with a surface area of 0.30 m^2 in an electrolytic cell with a current of 25 A? Given the density of chromium is 7.19 g/cm^3 .

$$m = \frac{AI t}{zF} \quad F = 9.65 \times 10^4 \quad (1 + 1 + 5 \text{ marks})$$

$$z = \text{mol of } \text{e}^-$$

- b) Michael Faraday was an English scientist who discovered the principles underlying electromagnetic and electrochemistry.

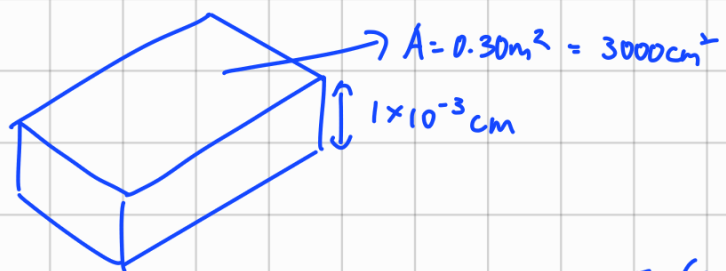
- i. State the Faraday's Second Law of electrolysis.
- ii. Predict how much of each of the following chemicals can be discharged by one Faraday of electricity.

Gold (III) chloride, AuCl_3

Hydrochloric acid, HCl

(1 + 2 marks)

(iii)



$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

$$\text{mass} = 7.19 (3000) (1 \times 10^{-3})$$

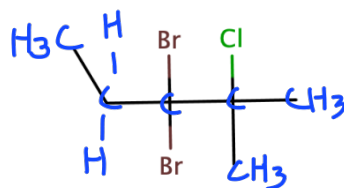
$$= 21.57 \text{ g}$$

$$21.57 = \frac{52(25)(t)}{3 \times 96500}$$

$$t = 4803 \text{ second}$$

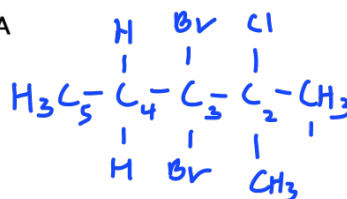
QUESTION 7 (10 MARKS)

- a) The secondary carbon in structure A has been successfully converted to tertiary carbon after undergone a few chemical reactions.



Structure A

For this new structure, you are required to:



i. Draw its structural formula.

ii. Provide its IUPAC name.

3,3 dibromo - 2-chloro - 2-methylpentane

iii. State its functional group

-C-C-

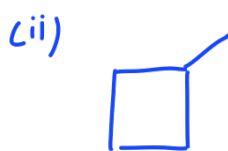
Carbon-carbon single bond for alkane

(1 + 2 + 1 marks)

- b) Structural isomers, also known as constitutional isomers are common in organic compounds. Draw **THREE (3)** structural isomers of C_5H_{10} that are cycloalkanes. State the name of each structural isomer.



cyclopentane



1-methylcyclobutane



1-ethylcyclopropane

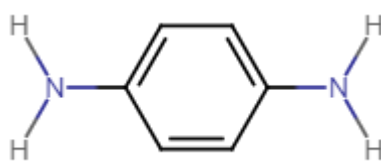
(6 marks)

QUESTION 8 (10 MARKS)

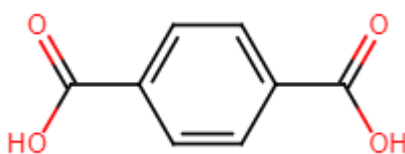
- a) 2-pentanol is an organic compound that is commonly used as a solvent and an intermediate in the synthesis of several potential anti-Alzheimer drugs and other chemicals.. Explain **ONE (1)** experimental procedure involved in testing 2-pentanol and the results.

(4 marks)

- b) Condensation polymerisation between benzene-1,4-diamine and benzene-1,4-dicarboxylic acid will produce a commercially available aramid fibre with excellent heat resistance and high stiffness which is used in bullet-proof vest, combat helmet, and as composites in aircraft structural parts.



benzene-1,4-diamine



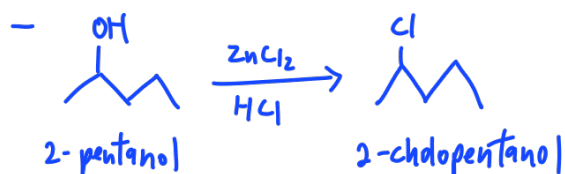
Benzene-1,4-dicarboxylic acid

- Explain how condensation polymerisation of these two monomers occur.
- Draw the repeating unit/monomer when the two monomers undergo condensation polymerisation.
- Identify the type of the polymer product, either homopolymer or copolymer. Justify your answer.

Q1) Lucas test

(2 + 2 + 2 marks)

- Lucas reagent added to 2-pentanol
- shake vigorously
- emulsion appears in 5-10 minutes



Appendix1 : THE PERIODIC TABLE

1	2	13	14	15	16	17	18
IA	IIA	IIIA	IVA	VA	VIA	VIIA	VIIIA
1 H 1.01	2 He 4.00	5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
3 Li 6.94	4 Be 9.01	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
11 Na 22.99	12 Mg 24.31	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
19 K 39.1	20 Ca 40.08	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.6	53 I 126.9	54 Xe 131.29
37 Rb 85.47	38 Sr 87.62	81 Tl 204.4	82 Pb 207.2	83 Bi 209	84 Po (209)	85 At (210)	86 Rn (222)
55 Cs 132.9	56 Ba 137.3	89 Ac[^] (227)	90 Th 232.0	91 Pa (231)	92 U 238.0	93 Np (237)	94 Pu (244)
87 Fr (223)	88 Ra (226)	101 La[*] 138.9	102 Ce 140.1	103 Pr 140.9	104 Nd 144.2	105 Pm (145)	106 Sm 150.4
		107 Eu 152.0	108 Gd 157.3	109 Tb 158.9	110 Dy 162.5	111 Ho 164.9	112 Er 167.3
		113 Yb 173.0	114 Lu 175.0	115 Yt 176.9	116 La 178.9	117 Ce 180.9	118 Pr 182.9
		119 Sm 183.9	120 Eu 186.2	121 Gd 188.9	122 Tb 190.9	123 Dy 193.2	124 Ho 195.1
		125 Er 197.0	126 Tm 198.9	127 Yb 200.0	128 Lu 200.9	129 La 201.1	130 Ce 201.2
		131 Pr 201.3	132 Nd 201.4	133 Pm (202)	134 Sm 202.1	135 Eu 203.4	136 Gd 204.4
		137 Tm 208.9	138 Yb 209.0	139 Lu 209.1	140 La 209.2	141 Ce 209.3	142 Pr 209.4
		143 Pr 209.5	144 Nd 209.6	145 Pm (209)	146 Sm 209.7	147 Eu 209.8	148 Gd 209.9
		149 Sm 209.9	150 Eu 210.0	151 Gd 210.1	152 Tb 210.2	153 Dy 210.3	154 Ho 210.4
		155 Er 210.5	156 Tm 210.6	157 Yb 210.7	158 Lu 210.8	159 La 210.9	160 Ce 211.0
		161 Pr 211.1	162 Nd 211.2	163 Pm (211)	164 Sm 211.3	165 Eu 211.4	166 Gd 211.5
		167 Tm 211.6	168 Yb 211.7	169 Lu 211.8	170 La 211.9	171 Ce 212.0	172 Pr 212.1
		173 Pr 212.2	174 Nd 212.3	175 Pm (212)	176 Sm 212.4	177 Eu 212.5	178 Gd 212.6
		179 Sm 212.6	180 Eu 212.7	181 Gd 212.8	182 Tb 212.9	183 Dy 213.0	184 Ho 213.1
		185 Er 213.2	186 Tm 213.3	187 Yb 213.4	188 Lu 213.5	189 La 213.6	190 Ce 213.7
		191 Pr 213.8	192 Nd 213.9	193 Pm (213)	194 Sm 214.0	195 Eu 214.1	196 Gd 214.2
		197 Tm 214.3	198 Yb 214.4	199 Lu 214.5	200 La 214.6	201 Ce 214.7	202 Pr 214.8
		203 Pr 214.9	204 Nd 215.0	205 Pm (215)	206 Sm 215.1	207 Eu 215.2	208 Gd 215.3
		209 Sm 215.4	210 Eu 215.5	211 Gd 215.6	212 Tb 215.7	213 Dy 215.8	214 Ho 215.9
		215 Er 216.0	216 Tm 216.1	217 Yb 216.2	218 Lu 216.3	219 La 216.4	220 Ce 216.5
		221 Pr 216.6	222 Nd 216.7	223 Pm (216)	224 Sm 216.8	225 Eu 216.9	226 Gd 217.0
		227 Tm 217.1	228 Yb 217.2	229 Lu 217.3	230 La 217.4	231 Ce 217.5	232 Pr 217.6
		233 Pr 217.7	234 Nd 217.8	235 Pm (217)	236 Sm 217.9	237 Eu 218.0	238 Gd 218.1
		239 Sm 218.2	240 Eu 218.3	241 Gd 218.4	242 Tb 218.5	243 Dy 218.6	244 Ho 218.7
		245 Er 218.8	246 Tm 218.9	247 Yb 219.0	248 Lu 219.1	249 La 219.2	250 Ce 219.3
		251 Pr 219.4	252 Nd 219.5	253 Pm (219)	254 Sm 219.6	255 Eu 219.7	256 Gd 219.8
		257 Tm 219.9	258 Yb 220.0	259 Lu 220.1	260 La 220.2	261 Ce 220.3	262 Pr 220.4
		263 Pr 220.5	264 Nd 220.6	265 Pm (220)	266 Sm 220.7	267 Eu 220.8	268 Gd 220.9
		269 Sm 221.0	270 Eu 221.1	271 Gd 221.2	272 Tb 221.3	273 Dy 221.4	274 Ho 221.5
		275 Er 221.6	276 Tm 221.7	277 Yb 221.8	278 Lu 221.9	279 La 222.0	280 Ce 222.1
		281 Pr 222.2	282 Nd 222.3	283 Pm (222)	284 Sm 222.4	285 Eu 222.5	286 Gd 222.6
		287 Tm 222.7	288 Yb 222.8	289 Lu 222.9	290 La 223.0	291 Ce 223.1	292 Pr 223.2
		293 Pr 223.3	294 Nd 223.4	295 Pm (223)	296 Sm 223.5	297 Eu 223.6	298 Gd 223.7
		299 Sm 223.8	300 Eu 223.9	301 Gd 224.0	302 Tb 224.1	303 Dy 224.2	304 Ho 224.3
		305 Er 224.4	306 Tm 224.5	307 Yb 224.6	308 Lu 224.7	309 La 224.8	310 Ce 224.9
		311 Pr 225.0	312 Nd 225.1	313 Pm (225)	314 Sm 225.2	315 Eu 225.3	316 Gd 225.4
		317 Tm 225.5	318 Yb 225.6	319 Lu 225.7	320 La 225.8	321 Ce 225.9	322 Pr 226.0
		323 Pr 226.1	324 Nd 226.2	325 Pm (226)	326 Sm 226.3	327 Eu 226.4	328 Gd 226.5
		329 Sm 226.6	330 Eu 226.7	331 Gd 226.8	332 Tb 226.9	333 Dy 227.0	334 Ho 227.1
		335 Er 227.2	336 Tm 227.3	337 Yb 227.4	338 Lu 227.5	339 La 227.6	340 Ce 227.7
		341 Pr 227.8	342 Nd 227.9	343 Pm (227)	344 Sm 228.0	345 Eu 228.1	346 Gd 228.2
		347 Tm 228.3	348 Yb 228.4	349 Lu 228.5	350 La 228.6	351 Ce 228.7	352 Pr 228.8
		353 Pr 228.9	354 Nd 229.0	355 Pm (229)	356 Sm 229.1	357 Eu 229.2	358 Gd 229.3
		359 Sm 229.4	360 Eu 229.5	361 Gd 229.6	362 Tb 229.7	363 Dy 229.8	364 Ho 229.9
		365 Er 230.0	366 Tm 230.1	367 Yb 230.2	368 Lu 230.3	369 La 230.4	370 Ce 230.5
		371 Pr 230.6	372 Nd 230.7	373 Pm (230)	374 Sm 230.8	375 Eu 230.9	376 Gd 231.0
		377 Tm 231.1	378 Yb 231.2	379 Lu 231.3	380 La 231.4	381 Ce 231.5	382 Pr 231.6
		383 Pr 231.7	384 Nd 231.8	385 Pm (231)	386 Sm 231.9	387 Eu 232.0	388 Gd 232.1
		389 Sm 232.2	390 Eu 232.3	391 Gd 232.4	392 Tb 232.5	393 Dy 232.6	394 Ho 232.7
		395 Er 232.8	396 Tm 232.9	397 Yb 233.0	398 Lu 233.1	399 La 233.2	400 Ce 233.3
		401 Pr 233.4	402 Nd 233.5	403 Pm (233)	404 Sm 233.6	405 Eu 233.7	406 Gd 233.8
		407 Tm 233.9	408 Yb 234.0	409 Lu 234.1	410 La 234.2	411 Ce 234.3	412 Pr 234.4
		413 Pr 234.5	414 Nd 234.6	415 Pm (234)	416 Sm 234.7	417 Eu 234.8	418 Gd 234.9
		419 Sm 235.0	420 Eu 235.1	421 Gd 235.2	422 Tb 235.3	423 Dy 235.4	424 Ho 235.5
		425 Er 235.6	426 Tm 235.7	427 Yb 235.8	428 Lu 235.9	429 La 236.0	430 Ce 236.1
		431 Pr 236.2	432 Nd 236.3	433 Pm (236)	434 Sm 236.4	435 Eu 236.5	436 Gd 236.6
		437 Tm 236.7	438 Yb 236.8	439 Lu 236.9	440 La 237.0	441 Ce 237.1	442 Pr 237.2
		443 Pr 237.3	444 Nd 237.4	445 Pm (237)	446 Sm 237.5	447 Eu 237.6	448 Gd 237.7
		449 Sm 237.8	450 Eu 237.9	451 Gd 238.0	452 Tb 238.1	453 Dy 238.2	454 Ho 238.3
		455 Er 238.4	456 Tm 238.5	457 Yb 238.6	458 Lu 238.7	459 La 238.8	460 Ce 238.9
		461 Pr 239.0	462 Nd 239.1	463 Pm (239)	464 Sm 239.2	465 Eu 239.3	466 Gd 239.4
		467 Tm 239.5	468 Yb 239.6	469 Lu 239.7	470 La 239.8	471 Ce 239.9	472 Pr 240.0
		473 Pr 240.1	474 Nd 240.2	475 Pm (240)	476 Sm 240.3	477 Eu 240.4	478 Gd 240.5
		479 Sm 240.6	480 Eu 240.7	481 Gd 240.8	482 Tb 240.9	483 Dy 241.0	484 Ho 241.1
		485 Er 241.2	486 Tm 241.3	487 Yb 241.4	488 Lu 241.5	489 La 241.6	490 Ce 241.7
		491 Pr 241.8	492 Nd 241.9	493 Pm (241)	494 Sm 242.0	495 Eu 242.1	496 Gd 242.2
		497 Tm 242.3	498 Yb 242.4	499 Lu 242.5	500 La 242.6	501 Ce 242.7	502 Pr 242.8
		503 Pr 242.9	504 Nd 243.0	505 Pm (243)	506 Sm 243.1	507 Eu 243.2	508 Gd 243.3
		509 Sm 243.4	510 Eu 243.5	511 Gd 243.6	512 Tb 243.7	513 Dy 243.8	514 Ho 243.9
		515 Er 244.0	516 Tm 244.1	517 Yb 244.2	518 Lu 244.3	519 La 244.4	520 Ce 244.5
		521 Pr 244.6	522 Nd 244.7	523 Pm (244)	524 Sm 244.8	525 Eu 244.9	526 Gd 245.0
		527 Tm 245.1	528 Yb 245.2	529 Lu 245.3	530 La 245.4	531 Ce 245.5	532 Pr 245.6
		533 Pr 245.7	534 Nd 245.8	535 Pm (245)	536 Sm 245.9	537	

Appendix 2 :LIST OF SELECTED CONSTANT VALUES

Speed of light in a vacuum	c	=	$3.0 \times 10^8 \text{ m s}^{-1}$
Avogadro's number	N_A	=	$6.02 \times 10^{23} \text{ mol}^{-1}$
Faraday constant	F	=	$9.65 \times 10^4 \text{ C mol}^{-1}$
Planck constant	h	=	$6.6256 \times 10^{-34} \text{ J s}$
Reduced Planck constant	$\hbar$	=	$1.054 \times 10^{-34} \text{ J s}$
Rydberg constant	R_H	=	$1.097 \times 10^7 \text{ m}^{-1}$ = $2.18 \times 10^{-14} \text{ J}$
Molar of gases constant	R	=	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$ = $0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$
Boltzmann constant	k	=	$1.3807 \times 10^{-27} \text{ J K}^{-1}$
Mass of proton	M_p	=	$1.672 \times 10^{-27} \text{ kg}$
Electronic Bohr magneton	m_B	=	$9.2741 \times 10^{-24} \text{ J T}^{-1}$
Nuclear Bohr magneton	b_N	=	$5.05 \times 10^{-27} \text{ J T}^{-1}$
Vapour pressure of water	P_{water}	=	23.8 torr
Electron charge	e^-	=	$1.602 \times 10^{-19} \text{ C}$

UNIT AND CONVERSION FACTOR

Energy	$1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2} = 1 \text{ N m} = 10^7 \text{ erg}$ $1 \text{ calorie} = 4.184 \text{ Joule}$ $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$ $1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$
Pressure	$1 \text{ atm} = 760 \text{ mm Hg} = 760 \text{ torr} = 101.325 \text{ kPa} = 101325 \text{ N m}^{-2}$

Appendix 3 : STANDARD REDUCTION POTENTIAL (E°_{Red})

$\text{PbO}_2(\text{s}) + \text{SO}_4^{2-}(\text{aq}) + 4\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{PbSO}_4(\text{s}) + 2\text{H}_2\text{O}(\ell)$	+1.69
$\text{MnO}_4^-(\text{aq}) + 8\text{H}^+(\text{aq}) + 5\text{e}^- \rightarrow \text{Mn}^{2+}(\text{aq}) + 4\text{H}_2\text{O}(\ell)$	+1.51
$\text{Au}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Au}(\text{s})$	+1.50
$\text{ClO}_4^-(\text{aq}) + 8\text{H}^+(\text{aq}) + 8\text{e}^- \rightarrow \text{Cl}^-(\text{aq}) + 4\text{H}_2\text{O}(\ell)$	+1.39
$\text{Cl}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{Cl}^-(\text{aq})$	+1.36
$\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14\text{H}^+(\text{aq}) + 6\text{e}^- \rightarrow 2\text{Cr}^{3+}(\text{aq}) + 7\text{H}_2\text{O}(\ell)$	+1.33
$2\text{HNO}_3(\text{aq}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow \text{N}_2\text{O}(\text{g}) + 3\text{H}_2\text{O}(\ell)$	+1.30
$\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}(\ell)$	+1.23
$\text{MnO}_2(\text{s}) + 4\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{Mn}^{2+}(\text{aq}) + 2\text{H}_2\text{O}(\ell)$	+1.22
$\text{Br}_2(\text{aq}) + 2\text{e}^- \rightarrow 2\text{Br}^-(\text{aq})$	+1.07
$\text{Hg}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Hg}(\ell)$	+0.85
$\text{ClO}^-(\text{aq}) + \text{H}_2\text{O}(\ell) + 2\text{e}^- \rightarrow \text{Cl}^-(\text{aq}) + 2\text{OH}^-(\text{aq})$	+0.84
$\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow \text{Ag}(\text{s})$	+0.80
$\text{NO}_3^-(\text{aq}) + 2\text{H}^+(\text{aq}) + \text{e}^- \rightarrow \text{NO}_2(\text{g}) + \text{H}_2\text{O}(\ell)$	+0.80
$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Fe}^{2+}(\text{aq})$	+0.77
$\text{O}_2(\text{g}) + 2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2\text{O}_2(\ell)$	+0.70
$\text{I}_2(\text{s}) + 2\text{e}^- \rightarrow 2\text{I}^-(\text{aq})$	+0.54
$\text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\ell) + 4\text{e}^- \rightarrow 4\text{OH}^-(\text{aq})$	+0.40
$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$	+0.34
$\text{SO}_4^{2-}(\text{aq}) + 4\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2\text{SO}_3(\text{aq}) + \text{H}_2\text{O}(\ell)$	+0.17
$\text{Sn}^{4+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Sn}^{2+}(\text{aq})$	+0.15
$\text{S}(\text{s}) + 2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2\text{S}(\text{aq})$	+0.14
$\text{AgBr}(\text{s}) + \text{e}^- \rightarrow \text{Ag}(\text{s}) + \text{Br}^-(\text{aq})$	+0.07
$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_{2(\text{g})}$	0.00
$\text{Pb}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Pb}(\text{s})$	-0.13
$\text{Sn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Sn}(\text{s})$	-0.14
$\text{AgI}(\text{s}) + \text{e}^- \rightarrow \text{Ag}(\text{s}) + \text{I}^-(\text{aq})$	-0.15
$\text{Ni}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Ni}(\text{s})$	-0.26
$\text{Co}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Co}(\text{s})$	-0.28
$\text{PbSO}_4(\text{s}) + 2\text{e}^- \rightarrow \text{Pb}(\text{s}) + \text{SO}_4^{2-}(\text{aq})$	-0.36
$\text{Se}(\text{s}) + 2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2\text{Se}(\text{aq})$	-0.40
$\text{Cd}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cd}(\text{s})$	-0.40
$\text{Cr}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Cr}^{2+}(\text{aq})$	-0.41
$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Fe}(\text{s})$	-0.45
$\text{NO}_2^-(\text{aq}) + \text{H}_2\text{O}(\ell) + \text{e}^- \rightarrow \text{NO}(\text{g}) + 2\text{OH}^-(\text{aq})$	-0.46
$\text{Ag}_2\text{S}(\text{s}) + 2\text{e}^- \rightarrow 2\text{Ag}(\text{s}) + \text{S}^{2-}(\text{aq})$	-0.69
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Zn}(\text{s})$	-0.76
$2\text{H}_2\text{O}(\ell) + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$	-0.83
$\text{Cr}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cr}(\text{s})$	-0.91
$\text{Se}(\text{s}) + 2\text{e}^- \rightarrow \text{Se}^{2-}(\text{aq})$	-0.92
$\text{SO}_4^{2-}(\text{aq}) + \text{H}_2\text{O}(\ell) + 2\text{e}^- \rightarrow \text{SO}_3^{2-}(\text{aq}) + 2\text{OH}^-(\text{aq})$	-0.93
$\text{Al}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Al}(\text{s})$	-1.66
$\text{Mg}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Mg}(\text{s})$	-2.37
$\text{Na}^+(\text{aq}) + \text{e}^- \rightarrow \text{Na}(\text{s})$	-2.71
$\text{Ca}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Ca}(\text{s})$	-2.87
$\text{Ba}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Ba}(\text{s})$	-2.91
$\text{Li}^+(\text{aq}) + \text{e}^- \rightarrow \text{Li}(\text{s})$	-3.04